



University of Piraeus

**School of information and Communication Technologies
Department of Digital Systems**

Bachelors Thesis

Event Reconstruction with Graph Neural Networks in ANNIE

Emmanouil Giannakis

**Supervising Professor
George Vouros**

Piraeus

May 2025

Bachelors Thesis

Event Reconstruction using Graph Neural Networks in ANNIE

Emmanouil Giannakis

A.M.: E21022

ΠΕΡΙΛΗΨΗ

Τα νετρίνα συγκαταλέγονται μεταξύ των πιο άφθονων στοιχειωδών σωματιδίων στο Σύμπαν, ωστόσο παραμένουν από τα λιγότερο κατανοητά. Η πειραματική τους μελέτη είναι ιδιαίτερα απαιτητική λόγω της εξαιρετικά μικρής διατομής αλληλεπίδρασής τους με την ύλη και της πολυπλοκότητας των αλληλεπιδράσεων με τους ατομικούς πυρήνες. Παρά τις δυσκολίες αυτές, η διερεύνηση των ιδιοτήτων τους θεωρείται κρίσιμη για την κατανόηση θεμελιωδών ζητημάτων της σωματιδιακής φυσικής και της κοσμολογίας, όπως η προέλευση της ασυμμετρίας ύλης–αντιύλης και οι μηχανισμοί που διέπουν τις ταλαντώσεις τους μεταξύ διαφορετικών ιδιοκαταστάσεων γεύσης.

Η περιορισμένη αλληλεπιδραστικότητά τους καθιστά αναγκαία την ανάπτυξη μεγάλων ανιχνευτικών συστημάτων και προηγμένων μεθοδολογιών για την ανίχνευση και την ακριβή ανακατασκευή των γεγονότων. Τα τελευταία χρόνια, οι μέθοδοι βαθιάς μάθησης έχουν αναδειχθεί ως ισχυρά εργαλεία στην ανάλυση δεδομένων σωματιδιακής φυσικής, παρέχοντας βελτιωμένες δυνατότητες εξαγωγής πληροφορίας από πολύπλοκες και αραιές κατανομές δεδομένων. Στο πλαίσιο αυτό, τα Νευρωνικά Δίκτυα Γράφων (Graph Neural Networks – GNNs) έχουν επιδείξει ιδιαίτερη αποτελεσματικότητα σε πειράματα ανίχνευσης νετρίνων, όπως τα IceCube και KM3NeT, λόγω της δυνατότητάς τους να επεξεργάζονται εγγενώς μη πλεγματικές και μη ομοιόμορφες δομές δεδομένων.

Η παρούσα εργασία αξιοποιεί το GraphNeT framework, που αναπτύχθηκε από το πείραμα IceCube, εφαρμόζοντας την αρχιτεκτονική DynEdge και υποστηρίζοντας πολλαπλές μορφές δεδομένων. Η επιλογή των GNNs βασίζεται στην ικανότητά τους να προσαρμόζονται σε ακανόνιστες και αραιές κατανομές γεγονότων, χαρακτηριστικό που είναι καίριας σημασίας για την ανάλυση δεδομένων νετρίνων.

ΘΕΜΑΤΙΚΗ ΠΕΡΙΟΧΗ: Βαθιά Μάθηση σε εφαρμογές Σωματιδιακής Φυσικής

ΛΕΞΕΙΣ ΚΛΕΙΔΙΑ: Νετρίνα, Νευρωνικά Δίκτυα Γράφων, ANNIE, ενέργεια, κορυφή

ABSTRACT

Neutrinos are among the most abundant elementary particles in the Universe, yet they remain among the least understood. Their experimental study is particularly challenging due to their extremely small interaction cross section with matter and the complexity of their interactions with atomic nuclei. Nevertheless, they are considered crucial for addressing fundamental questions in particle physics and cosmology, such as the origin of matter–antimatter asymmetry and the mechanisms underlying their oscillations between different flavor eigenstates. These properties make neutrinos a central focus of modern scientific research.

Detecting and accurately reconstructing such interactions necessitates the development of large-scale detection systems and advanced reconstruction methodologies. In recent years, deep learning has emerged as a powerful analytical framework in particle physics, particularly in neutrino experiments, providing enhanced capabilities for the analysis and interpretation of complex datasets. Within this context, neural networks have been investigated as a means to address the challenges associated with neutrino detection and interaction reconstruction.

Specifically, Graph Neural Networks (GNNs) have demonstrated strong performance in similar experiments, such as IceCube and KM3NeT, and were selected for their ability to natively represent the irregular, non-grid structure and sparse nature of neutrino event data. This study employed the GraphNeT framework, developed by the IceCube collaboration, implementing the DynEdge architecture and supporting multiple data formats.

SUBJECT AREA: Deep Learning in Particle Physics

KEYWORDS: Neutrinos, Graph Neural Networks, ANNIE, energy, vertex, LAPPD

Acknowledgements

I would like to express my deepest gratitude to the Institute of Nuclear and Particle Physics at NCSR Demokritos for welcoming me into their research environment and providing the resources and support that made this thesis possible. It has been a privilege to work among such dedicated scientists whose passion for discovery is truly inspiring.

I am especially grateful to Dr. Evangelia Drakopoulou for her exceptional guidance, mentorship, and constant support throughout the course of this project. Her deep expertise in particle physics, combined with her patience and encouragement, were instrumental not only in navigating the scientific challenges of this thesis but also in my personal growth as a researcher. Her readiness to help at every stage, no matter how complex the problem or how simple the question, made a lasting impression on me and shaped my experience in the most positive way.

Working under her supervision has been one of the most valuable and formative experiences of my academic journey. For her unwavering support and for setting a standard of excellence I will always strive to follow, I am sincerely thankful.

To my family and friends, whose unwavering support and encouragement have carried me through every challenge.

To my parents, for their love, sacrifices, and belief in me even when I doubted myself.

To my friends, for keeping me grounded and proving that good company is the best antidote to stress.

And most especially, to Evan and Eugenia — thank you for your constant presence, kindness, and for making this journey brighter.

Contents

Acknowledgements.....	5
1. Introduction	9
2. Neutrinos.....	11
2.1 Neutrino Physics	11
2.1.1 The Standard Model of Particle Physics	12
2.1.2 Neutrino Interactions.....	14
2.2 Detection Mechanisms.....	17
2.2.1 Water Cherenkov Neutrino Detectors	18
2.2.2 Principles of Cherenkov Radiation	19
2.2.2.1 Emission Condition and Angle	20
2.2.2.2 Spectral Distribution	20
2.2.2.3 Energy Threshold	21
2.2.2.4 Coherence and Timing	21
3. The Accelerator Neutrino Neutron Interaction Experiment	23
3.1 Booster Neutrino Beam (BNB)	24
3.2 Neutron Multiplicity in ANNIE	27
3.3 Water Tank.....	28
3.4 PMT and LAPPDs.....	31
3.4.1 Photomultiplier Tube (PMT)	31
3.4.2 LARGE AREA PICOSECOND PHOTODETECTORS (LAPPD)	32
3.5 Muon Range Detector (MRD).....	34
4. Deep Learning and Graph Neural Networks.....	36
4.1 Deep Learning.....	36
4.1.1 What is Deep Learning.....	36
4.1.2 Architecture of Deep Neural Networks	38
4.1.3 Training Deep Learning Models.....	39
4.1.4 Loss Functions and Backpropagation.....	40
4.2 Graph Neural Networks	41
4.2.1 Event Graph Representation.....	42
4.2.2 Edge Convolution	44
4.2.3 DynEdge architecture.....	45

5. Data Collection and Preprocessing	47
5.1 Data Collection Software in ANNIE	47
5.1.1 ANNIE Monte Carlo Simulations	47
5.1.2 Data Acquisition System.....	48
5.2 Datasets and Data Preprocessing	50
5.2.1 Data Formats	50
5.2.2 Conversion from ROOT to HDF5, Parquet and SQLite.....	51
5.2.3 Data weighting and Filtering	52
6. Energy Reconstruction using Graph Neural Networks.....	54
6.1 Integrating ANNIE Deep Learning data Format in GraphNeT	54
6.2 Computational Environment.....	55
6.2.1 Loss Function Selection: Log-Cosh	56
6.4 Results.....	58
6.4.1 Parquet.....	58
6.4.2 SQLite	65
6.4.3 SQLite vs Parquet.....	69
6.5 Conclusion and Outlook.....	70
ABBREVIATIONS - ACRONYMS - ACRONYMS.....	72
Bibliography	73